Joshua Aney

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Education

Montana State University Bozeman, MT
Bachelor of Science, Computer Science, GPA: 3.96 *May 2026*

Research Experience

Research Technologist June 2026 – Present
CaMP Lab (Cybernetics and Motor Physiology), Mayo Clinic *Rochester, MN*

- Previously Summer Research Intern in the same lab, 2024 and 2025.
- Conducted algorithmic analysis on human sEEG data to study how the brain responds to auditory stimuli.
- Implemented signal processing pipelines including noise reduction, baseline correction, and artifact rejection.
- Applied and modified Canonical Response Parameterization to allow for better temporal feature extraction when applied to auditory evoked responses.
- Created a new method for dynamic brain network identification using evoked responses.
- Leading development of a reference and preprocessing guide for general sEEG data analysis.

Undergraduate Research Assistant January 2026 – June 2026
HACR Lab (Harnessing Automation in Cybersecurity Reasoning), Montana State University *Bozeman, MT*

- Built pipelines to produce control flow graphs from compiled binary packages for code embedding generation.
- Designed Graph Neural Network architectures using PyTorch based on recent research publications.
- Ran machine learning workloads on Tempest, Montana State University's high performance computing system.

Teaching & Curriculum Development

Independent Study – CSCI 491-7 Hypermedia Systems Fall 2025
Montana State University *Bozeman, MT*

- Collaborated with a professor and a team of three students to design and develop a semester-long project template for an advanced web development course teaching a hypermedia-first approach using server-side rendering with HTMX.
- Implemented core website functionality, including the MVC controller layer, authentication and authorization systems, and multi-step business workflows, for a veterinary clinic management system.
- Designed a database schema with 11 interconnected tables, including polymorphic relationships and conversation-based messaging, to teach role-based access control, pagination, and transactional integrity.
- Structured a progressive assignment sequence so students could implement new controllers by following established patterns in the codebase.

Publications & Presentations

Papers in Preparation

- Aney, J., et al. *Response Lag Correlation: Identifying response networks through evoked response lags.*
- Aney, J., et al. *A Reference and Preprocessing Guide for sEEG Data.*

Conference Abstracts

- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects.* BRAIN Initiative Conference, 2026.
- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects.* GATTLE, Mayo Clinic, 2026.

Posters

- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects.* Poster presented at BRAIN Initiative Conference, 2026.

Technical Projects

Machine Learning and AI Models from Scratch 2024-2026

- Implemented the following machine learning algorithms from scratch: Naïve Bayes, K-Nearest Neighbors, K-Means, and DBSCAN, Multi-Layer-Perceptron, Q-Learning, SARSA, and Value Iteration
- Implemented a First-Order-Logic engine to perform inference on knowledge bases using resolution.
- Implemented the following optimization algorithms from scratch: Backpropagation, Genetic Algorithm, Differential Evolution, Particle Swarm Optimization, and Simulated Annealing

Prize-Collecting TSP Solver 2026

- Formulated the Prize-Collecting Traveling Salesman Problem as a mixed-integer linear program and solved it exactly using IBM CPLEX.
- Implemented a genetic algorithm as an alternative heuristic solver and compared solution quality and runtime against the exact CPLEX formulation.

Bozeman Bike Trails Application 2024

- Developed a web application to track user-ridden bike trail GPS data

Hand Tracking / ASL Alphabet Detection 2024

- Created a pipeline to collect data, train a dense neural network, and provide real time predictions for video inputs.

Honors & Awards

President's List: Spring 2022; Spring 2023, Fall 2023; Spring 2024; Spring 2025, Fall 2025; Spring 2026, Fall 2026
Dean's List: Fall 2022; Fall 2024

Other Activities

President August 2023 – May 2026
Montana State University Tennis Club *Bozeman, MT*

Section Leader April 2025 – January 2026
Montana State University Drumline *Bozeman, MT*

Tennis Coach Summers 2023, 2025
Rochester Tennis Connection *Rochester, MN*

Substitute Tennis Coach 2022 – 2026
Bobcat Anderson Tennis Center, Montana State University *Bozeman, MT*